Part 5. Interaction Planning

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Principles

Interaction lets Natalia choose what to see, but every action has a cognitive cost (course 07). We therefore added only interactions that serve her seven questions, followed Shneiderman's mantra ("overview first, zoom and filter, then details on demand") and gave each one a keyboard equivalent. Both screens open with data already visible: both hotels, the whole period and, on the detail screen, Portugal selected. Zoom, pan and drag were rejected because the line and bar charts show at most 26 months or 11 country groups, and the map is only a small secondary view.

#	Interaction	Course type	Mantra step
I1	Hotel filter	Manipulation: filter	Zoom and filter
I2	Year filter	Manipulation: filter	Zoom and filter
I3	Filters kept between screens	Exploration: navigation	Overview first
I4	Reset filters	Manipulation: filter	Zoom and filter
I5	Linked month highlight	Problem solving: linked views	Details on demand
I6	Tooltips	Problem solving: details on demand	Details on demand
I7	Select a country	Problem solving: linked views	Details on demand
I8	Show more	Problem solving: overview and detail	Details on demand
I9	Sort toggle	Manipulation: order	Zoom and filter

Interactions between charts

The hotel and year filters (I1, I2) act on all KPIs and charts of a screen, and the link to the other screen carries them (I3), so Natalia does not have to remember and repeat her choice. The year labels name the partial years, "2015 (Jul–Dec)" and "2017 (Jan–Aug)", so they are not compared with 2016 by mistake. A reset button (I4) is always visible. On the main screen, pointing at a month or moving to it with the arrow keys draws a rule at that month in the three charts (I5), relating guests, cancellations and price of the same month. On the detail screen, selecting a country with a click, the arrow keys or the Previous and Next buttons (I7) updates its season line, the KPIs and the hidden charts; the selected bar gets an outline and its name and value are written beside the chart, so colour is not the only cue.

Interactions inside each chart

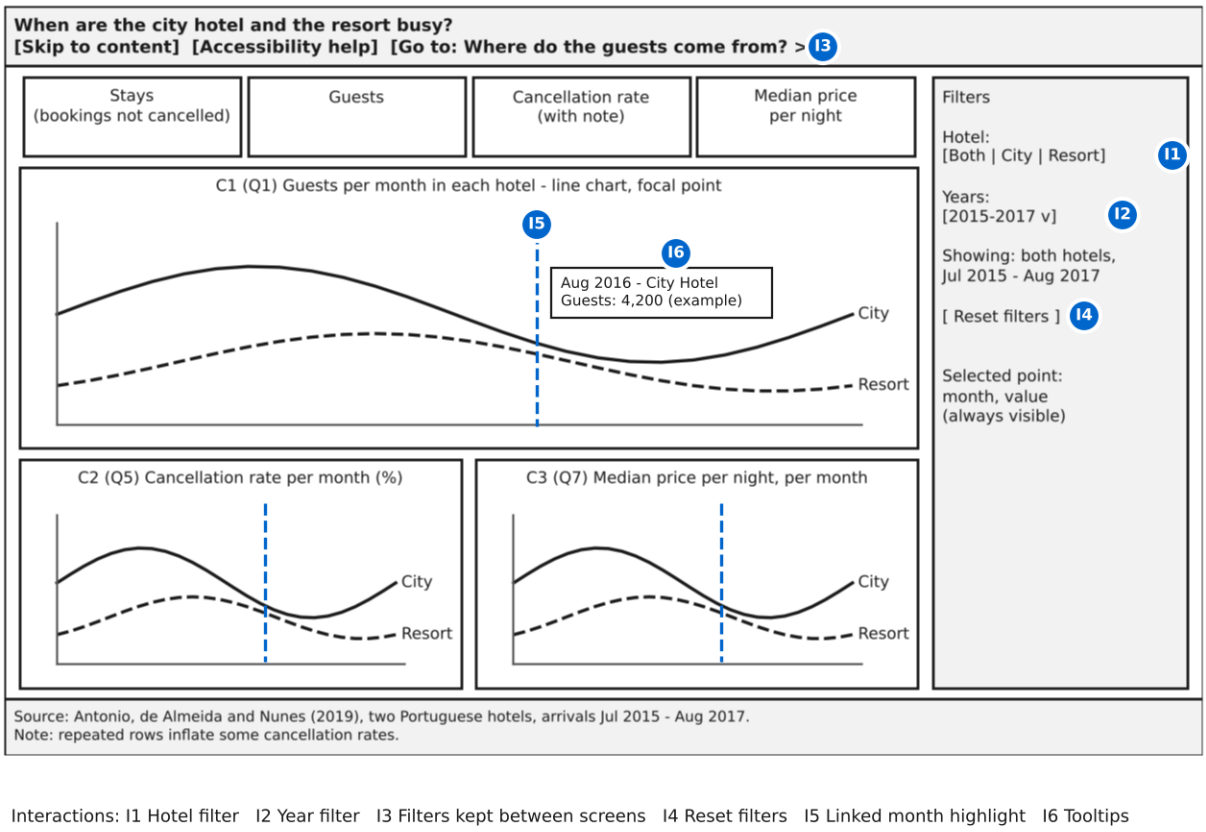
Tooltips (I6) give exact formatted values with units on every chart, also with Enter or Space on a focused point. The sort toggle (I9) reorders the country bars by guests or alphabetically. "Show more" (I8) reveals the length of stay, booking advance and map, keeping at most three charts visible.

Feasibility

Every chart feature was built in a small Vega-Altair test and compiled to SVG. Because each chart is embedded separately, with its own alternative text, selections are passed between charts with a few lines of JavaScript, built and tested in Part 6. Figures 1 and 2 show where each interaction appears.

Figures

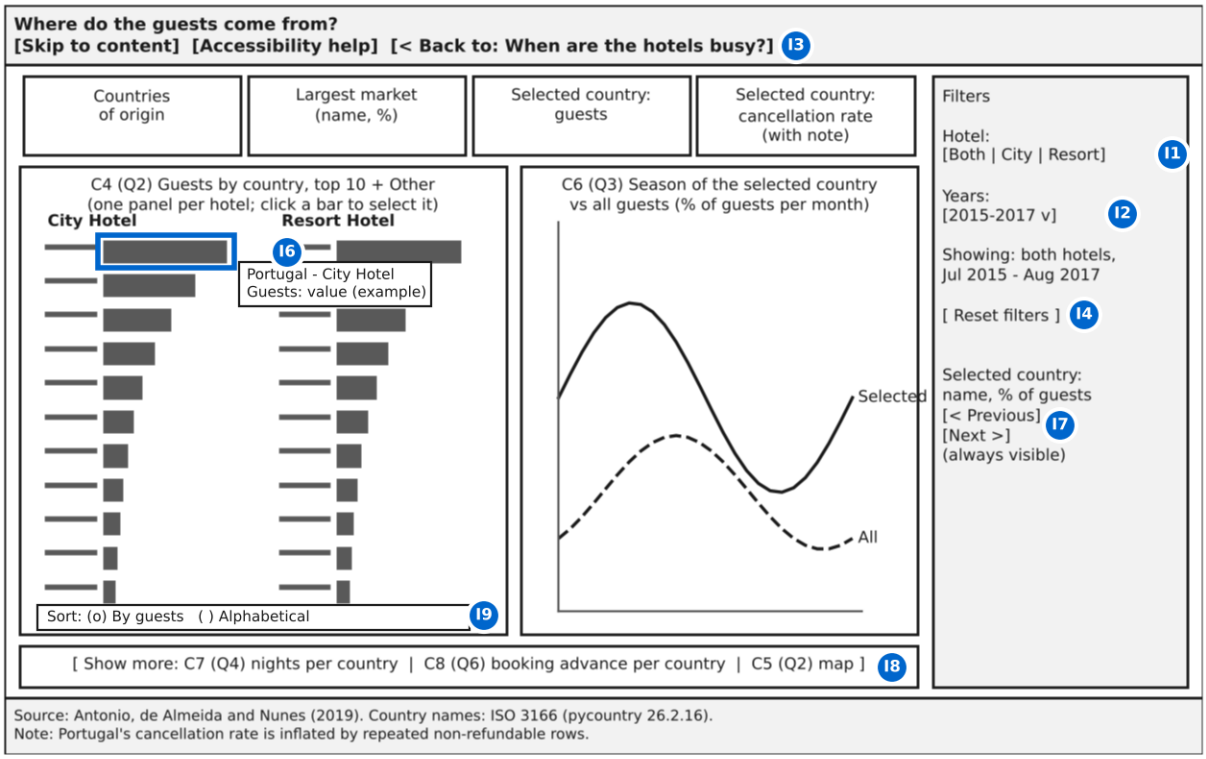
Prototype with interactions: main screen



Interactions: I1 Hotel filter I2 Year filter I3 Filters kept between screens I4 Reset filters I5 Linked month highlight I6 Tooltips

Figure 1. Main screen with interactions I1 to I6. Source: our design; code: prototype_interactions.ipynb.

Prototype with interactions: detail screen



Interactions: I1 Hotel filter I2 Year filter I3 Filters kept between screens I4 Reset filters I6 Tooltips I7 Select a country I8 Show more I9 Sort toggle

Figure 2. Detail screen with interactions I1 to I4 and I6 to I9. Source: as Figure 1.